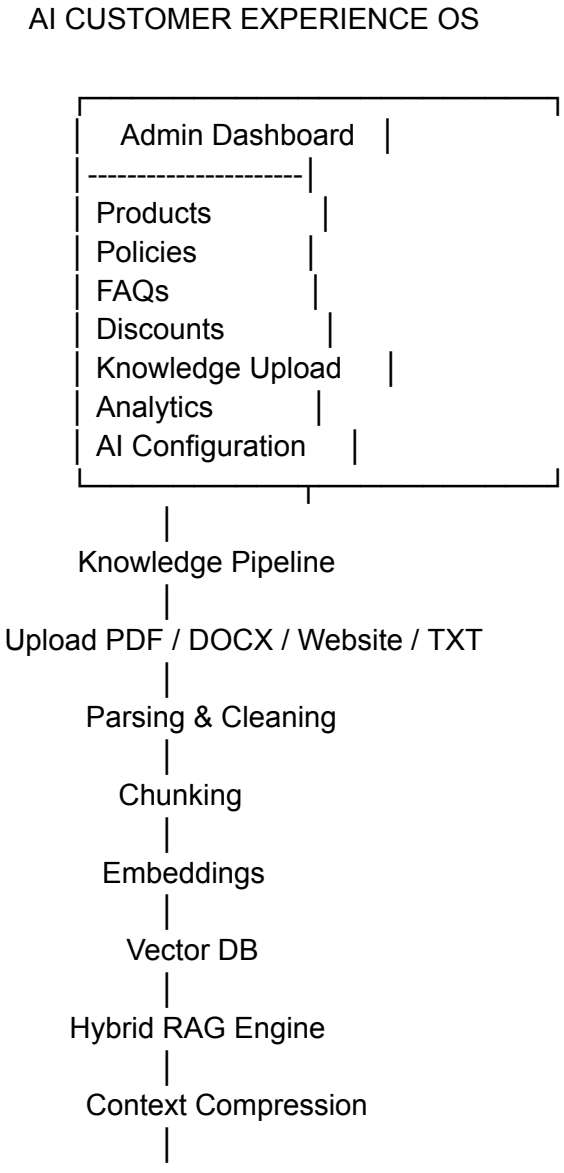
CortexCX

AI Customer Experience Operating System for D2C Businesses

An autonomous AI-powered Customer Experience Operating System designed for D2C businesses, combining Local LLMs, Advanced RAG, customer memory, AI planning, tool orchestration, human handoff, and multi-channel communication.

System Architecture



Cross Encoder Reranking

Communication Channels

• Website Chat
• Telegram Bot

Future Extensions

• WhatsApp
• Email
• Instagram

Communication Layer

Conversation Manager

Memory Engine

• Short-Term Memory
• Long-Term Memory
• Customer Profile
• Conversation History

Intent Detection

AI Planner



Tool Calling Advanced RAG

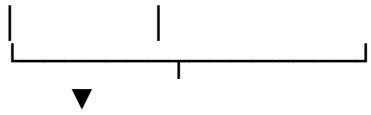
Order Tool Hybrid Search

Coupons Metadata Filtering

Inventory BM25 Retrieval

Policies Cross Encoder Reranker

Knowledge Base Context Compression



Local LLM (Ollama)



Confidence Evaluation

confidence > threshold

AI Response

confidence < threshold



Human Handoff



Agent Assist

• Suggested Reply
• Relevant Policies

• Conversation Summary

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Learning Engine

▼

Analytics Dashboard

Communication Flow

Telegram User

|

Telegram Bot API

|

Webhook

|

FastAPI Backend

|

Conversation Manager

|

Memory Engine

|

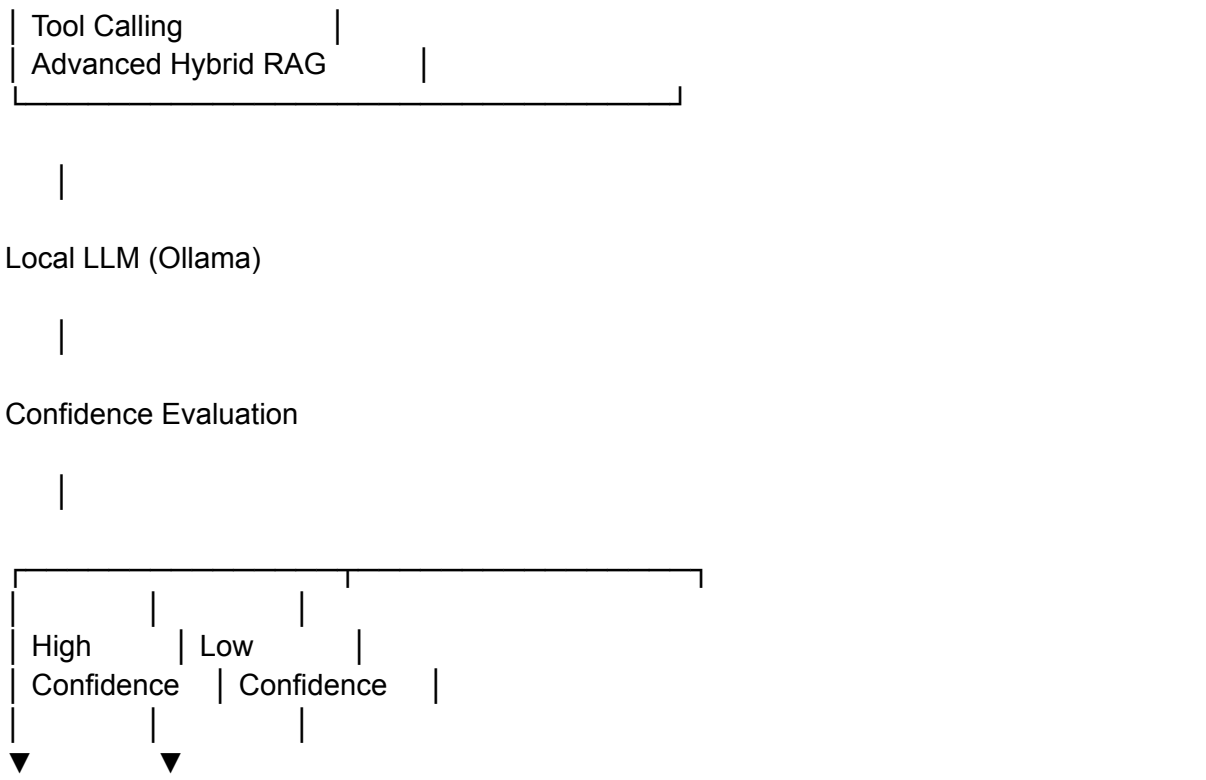
Intent Detection

|

AI Planner

|

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Knowledge Pipeline



Document Parsing

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Cleaning & Normalization

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Semantic Chunking

|

Embedding Generation

|

Vector Database Storage

|

Hybrid Retrieval

|

Metadata Filtering

|

Cross Encoder Reranking

|

Context Compression

|

Retrieved Context

|

Local LLM

AI Decision Flow

Customer Query

|

Conversation Manager

|

Memory Retrieval

|

Intent Detection

|

Task Planning

|

Needs Tool?

|

———————————

Yes

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Tool Selection

↓

Execute Tool

↓

Collect Results

———————————

No

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Knowledge Retrieval

↓

Advanced Hybrid RAG

↓

Context Assembly

↓

Local LLM

↓

Confidence Evaluation

↓

Response Generation

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Customer

Human Handoff Flow

Customer Question

|

AI Processing

|

Confidence Evaluation

|

Confidence Below Threshold

|

Escalation

|

Admin Dashboard

|

Agent Assist

• Suggested Response
• Retrieved Knowledge
• Relevant Policies
• Conversation Summary

|

Human Response

|

Customer

|

Learning Engine

Learning Pipeline

Human Corrections

|

Verified Responses

|

Knowledge Improvement

|

Update Retrieval Dataset

|

Improve Future Responses

Explainable AI Pipeline

Every AI-generated response includes:

- Confidence Score
- Retrieved Knowledge Sources
- Policies Referenced
- Documents Used
- Reasoning Summary

Core Modules

Module 1 – Admin Portal

- Dashboard
- Analytics
- Products
- Customers
- Knowledge Base
- Policies
- Coupons
- AI Configuration

Module 2 – Knowledge Builder

- PDF Upload
- DOCX Upload
- TXT Upload
- Website Import
- Automatic Parsing
- Semantic Chunking
- Embedding Generation
- Vector Database Storage

Module 3 – Advanced RAG Engine

- Hybrid Search
- Dense Retrieval

- BM25 Retrieval
- Metadata Filtering
- Cross Encoder Reranking
- Context Compression
- Citation Generation

Module 4 – Memory Engine

- Short-Term Memory
- Long-Term Memory
- Customer Profiles
- Conversation History
- Episodic Memory

Module 5 – AI Planner

Customer Query

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Intent Detection

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Task Planning

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Tool Selection

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Knowledge Retrieval

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Local LLM

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Response

Module 6 – Tool Calling

Mock APIs:

- Product Search
- Order Tracking
- Inventory Lookup
- Coupon Generation
- Refund Policy
- Knowledge Search

Module 7 – Explainable AI

Each response provides:

- Confidence Score
- Retrieved Sources
- Policies Used
- Supporting Documents
- Reasoning Summary

Module 8 – Human Handoff

Low Confidence

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Admin Dashboard

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Human Agent

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Continue Conversation

Module 9 – Learning Engine

Human Corrections

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Knowledge Update

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Retrieval Improvement

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Better Future Responses

Module 10 – Telegram Integration

Telegram User

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Telegram Bot API

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FastAPI Backend

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Conversation Manager

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AI Customer Experience Engine

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Telegram Response

Database Design (Channel Independent)

Users

Field	Description
id	Unique User ID

platform	Communication Platform
platform_user_id	Platform-specific User ID
username	Username
display_name	Customer Name
language	Preferred Language

Supported Communication Channels

Current

- Telegram Bot
- Website Chat

Planned Extensions

- WhatsApp
- Email
- Instagram
- Facebook Messenger

Technology Stack

Backend

- FastAPI
- Python

AI

- Ollama (Local LLM)
- LangChain / LlamaIndex
- Advanced Hybrid RAG

Database

- PostgreSQL
- Vector Database (ChromaDB / FAISS)

Frontend

- React
- Tailwind CSS

Communication

- Telegram Bot API
- Webhooks

Deployment

- Docker
- Nginx

Key Features

- Autonomous AI Customer Support
- AI Sales Assistance
- Local LLM Inference
- Advanced Hybrid RAG
- Long-Term Customer Memory
- AI Task Planning
- Tool Calling
- Explainable AI
- Human-in-the-Loop Support
- Knowledge Management
- Analytics Dashboard
- Multi-Channel Communication Architecture
- Modular & Scalable System Design